Department of Information Technology
Academic Year 2024 - 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT
Course Code & Title: CS3491 & Artificial Intelligence and Machine Learning
Name of the Faculty member (s): Mrs. G.Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / Support Vector Machine
- **Course Outcome:** CO3
- **Activity Chosen:** Reflection

- **Justification:**

  The topic of Support Vector Machines (SVM) is chosen for the reflection activity because SVM is a critical concept in machine learning and pattern recognition. Students are introduced to SVM as a supervised learning algorithm used for classification and regression tasks. It is essential for students to grasp how SVM works, including concepts such as hyperplanes, kernels, and margin maximization Reflection is a process that allows teachers to assess their teaching methods and consider ways to enhance their approach. In this reflection activity, students are asked to write down any questions or areas of confusion they have, enabling the instructor to address and clarify these concepts during class.
- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

  After explaining the concepts of Support Vector Machines (SVM) in class, students were encouraged to write down any doubts or areas where they needed further clarification. These doubts were collected by the instructor, and in the following class session, the instructor dedicated 10 minutes to address and clarify the students' concerns. Figure 1.a shows the doubts shared by students regarding the SVM topic, along with the responses provided by the instructor to clear those doubts.. This activity allowed students to review the SVM topic, express their uncertainties, and receive explanations. As a result, the students reported a clearer understanding of the SVM concepts.

- **CO – PO / PSO mapping:**

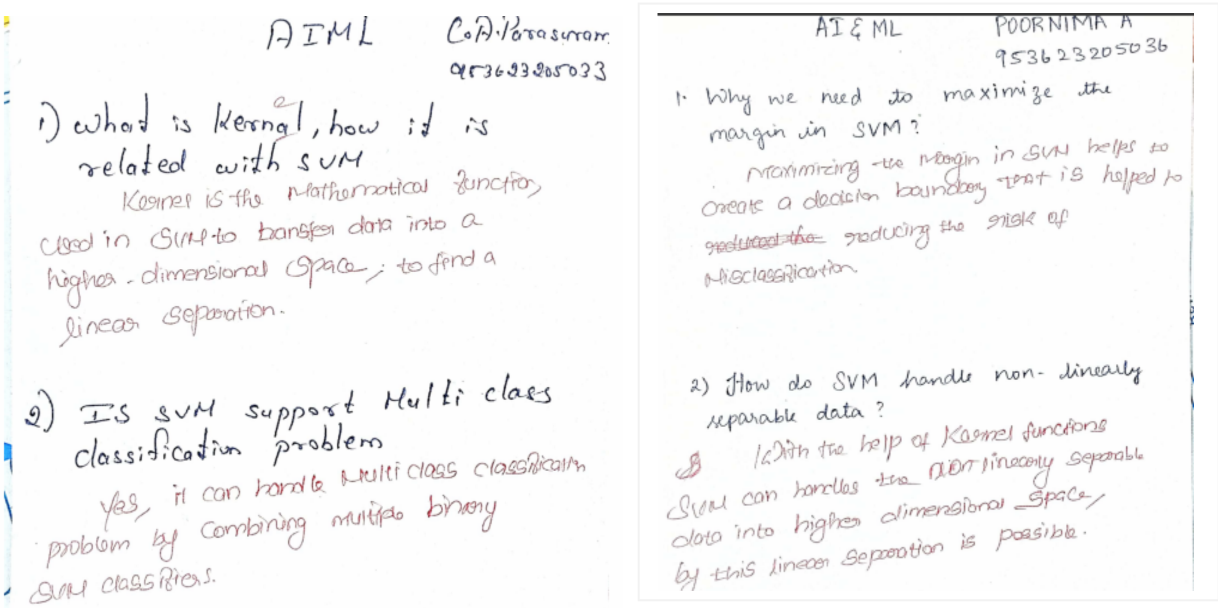
CO	PO1	PO3	PO11	PSO1	PSO2	PSO3
CO3	2	1	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO3	PO11	PSO1	PSO2	PSO3
CO3	2	1	2	1	1	1
Justification for correlation	Needs basic engineering knowledge for understand the concept of SVM.	The students can able to find solution for the complex ML problems using the SVM.	SVM helps students to work with projects like credit scoring, stock market prediction, and portfolio optimization.	SVM are used in many data science projects to solve complex computational problem	The students can apply the concept of SVM in research finding and provide reliable solution	The students can use this concept of SVM in software project development.

- **Images / Screenshot of the practice:**


AIML C.A.Parasuram
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1) what is Kernel, how it is related with SVM
Kernel is the Mathematical function used in SVM to transfer data into a higher-dimensional space, to find a linear separation.

2) Is SVM support Multi class classification problem
Yes, it can handle Multi class classification problem by combining multiple binary SVM classifiers.

AI & ML POORNIMA A
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1. Why we need to maximize the margin in SVM?
Maximizing the margin in SVM helps to create a decision boundary that is helped to ~~reduced the~~ reducing the risk of Misclassification.

2) How do SVM handle non-linearly separable data?
With the help of Kernel functions SVM can handles the non linearly separable data into higher dimensional space, by this linear separation is possible.

Figure 1. a Sample of Question shared by C. A. Parasuram and Poornima. A

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
    - The students felt that the activity give the opportunity to express their doubts openly, which helped them gain clarity on complex concepts.
    - They felt that the activity allowed them to understand the SVM topic better through discussions and additional explanations from the instructor.
    - Students also mentioned that the reflective activity enhanced their confidence in applying SVM concepts to practical problems.

- ***Benefit of the practice:***
    - The practice fostered collaborative learning, where students shared insights and learned from each other's perspectives.
    - Students reported that this reflective activity helped them retain the concepts better by revisiting and discussing key topics in detail.
- ***Challenges faced in implementation:***
    - Some students struggled to express their doubts clearly in proper English, resulting in a few unclear questions.
    - A few students were hesitant to share their doubts and not write anything.

References:

- https://apiar.org.au/wp- content/ uploads/ 2017/02/13_APJCECT _Feb_BRR798_ EDU-126-131.pdf

- https://www.skillsyouneed.com/ps/reflective-practice.html

- https://www.cambridge-community.org.uk/professional- development/ gswrp/ index.html

Signature of Faculty Member **HOD**